DLMS Testing Documentation

Testing Strategy

This document outlines comprehensive testing procedures for the Distance Learning Management System following the Waterfall SDLC methodology.

Test Environment

- **Operating System:** Windows/macOS/Linux
- **Python Version:** 3.7+
- **MySQL Version:** 5.7+
- **Browser:** Chrome 90+, Firefox 88+, Safari 14+

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1. Unit Testing

1.1 Authentication Module Tests

Test Case 1.1.1: Valid Login - Administrator

Objective: Verify administrator can log in with correct credentials
Prerequisites: Database is set up with default admin account
Test Steps:

1. Navigate to http://127.0.0.1:5000/login
2. Enter email: admin@dlms.com
3. Enter password: admin123
4. Click "Login" button

Expected Result:

- User is redirected to administrator dashboard
- Welcome message displays "Welcome, System Administrator!"
- Navigation shows role as "Administrator"

Status: ☐ Pass ☐ Fail

Test Case 1.1.2: Valid Login - Lecturer

Objective: Verify lecturer can log in with correct credentials
Test Steps:

1. Navigate to login page
2. Enter email: lecturer@dlms.com
3. Enter password: lecturer123
4. Click "Login"

Expected Result:

- Redirect to lecturer dashboard
- Displays lecturer-specific menu options

Status: ☐ Pass ☐ Fail

Test Case 1.1.3: Valid Login - Student

Objective: Verify student can log in with correct credentials
Test Steps:

1. Navigate to login page
2. Enter email: student@dlms.com
3. Enter password: student123
4. Click "Login"

Expected Result:

- Redirect to student dashboard
- Displays student-specific menu options

Status: ☐ Pass ☐ Fail

Test Case 1.1.4: Invalid Login - Wrong Password

Objective: Verify system rejects incorrect password
Test Steps:

1. Navigate to login page
2. Enter email: admin@dlms.com
3. Enter password: wrongpassword
4. Click "Login"

Expected Result:

- Login fails
- Error message: "Invalid email or password"
- User remains on login page

Status: ☐ Pass ☐ Fail

Test Case 1.1.5: Invalid Login - Non-existent User

Objective: Verify system rejects non-existent users
Test Steps:

1. Navigate to login page
2. Enter email: nonexistent@test.com
3. Enter password: anypassword
4. Click "Login"

Expected Result:

- Login fails
- Error message displayed
- User remains on login page

Status: ☐ Pass ☐ Fail

Test Case 1.1.6: Login - Deactivated Account

Objective: Verify deactivated users cannot log in
Test Steps:

1. Admin deactivates a user account
2. Attempt to login with deactivated account credentials

Expected Result:

- Login fails
- Message: "Your account has been deactivated"

Status: ☐ Pass ☐ Fail

Test Case 1.1.7: Logout Functionality

Objective: Verify logout clears session
Test Steps:

1. Log in as any user
2. Click "Logout" button
3. Attempt to access dashboard URL directly

Expected Result:

- User is logged out
- Session is cleared
- Redirect to login page with message
- Direct URL access denied

Status: ☐ Pass ☐ Fail

1.2 Role-Based Access Control Tests

Test Case 1.2.1: Student Access Restriction

Objective: Verify students cannot access admin/lecturer pages
Test Steps:

1. Log in as student
2. Try to access /admin/dashboard directly
3. Try to access /lecturer/dashboard directly

Expected Result:

- Access denied
- Redirect to appropriate page
- Error message displayed

Status: ☐ Pass ☐ Fail

Test Case 1.2.2: Lecturer Access Restriction

Objective: Verify lecturers cannot access admin pages
Test Steps:

1. Log in as lecturer
2. Try to access /admin/users directly

Expected Result:

- Access denied
- Appropriate error message

Status: ☐ Pass ☐ Fail

Test Case 1.2.3: Unauthenticated Access

Objective: Verify unauthenticated users cannot access protected pages
Test Steps:

1. Ensure logged out
2. Try to access /admin/dashboard
3. Try to access /lecturer/courses
4. Try to access /student/courses

Expected Result:

- All attempts redirect to login page
- Message: "Please log in to access this page"

Status: ☐ Pass ☐ Fail

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2. Administrator Module Tests

Test Case 2.1: Create User - Valid Data

Objective: Verify admin can create new users
Test Steps:

1. Log in as administrator
2. Navigate to "Create User"
3. Enter full name: "Test Student"
4. Enter email: "teststudent@test.com"

5. Enter password: "password123"
6. Select role: "Student"
7. Click "Create User"

Expected Result:

- User created successfully
- Success message displayed
- User appears in user list
- Can log in with new credentials

Status: ☐ Pass ☐ Fail

Test Case 2.2: Create User - Duplicate Email

Objective: Verify system prevents duplicate emails
Test Steps:

1. Attempt to create user with existing email

Expected Result:

- Creation fails
- Error message: "Email already exists"

Status: ☐ Pass ☐ Fail

Test Case 2.3: Create User - Missing Fields

Objective: Verify form validation
Test Steps:

1. Navigate to create user form
2. Leave required fields empty
3. Attempt to submit

Expected Result:

- Form validation prevents submission
- Required field indicators appear

Status: ☐ Pass ☐ Fail

Test Case 2.4: Toggle User Status

Objective: Verify admin can activate/deactivate users
Test Steps:

1. View user list
2. Click "Deactivate" on active user
3. Verify status changes to "Inactive"
4. Click "Activate" on inactive user
5. Verify status changes to "Active"

Expected Result:

- Status toggles correctly
- Badge color changes
- Confirmation message displayed

Status: ☐ Pass ☐ Fail

Test Case 2.5: View All Courses

Objective: Verify admin can view all courses
Test Steps:

1. Navigate to "View All Courses"
2. Verify course list displays

Expected Result:

- All courses visible
- Shows course details and lecture count
- Displays lecturer names

Status: ☐ Pass ☐ Fail

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3. Lecturer Module Tests

Test Case 3.1: Create Course - Valid Data

Objective: Verify lecturer can create courses
Test Steps:

1. Log in as lecturer
2. Navigate to "Create Course"
3. Enter title: "Test Course"
4. Enter description: "This is a test course"
5. Click "Create Course"

Expected Result:

- Course created successfully
- Success message displayed
- Course appears in lecturer's course list

Status: ☐ Pass ☐ Fail

Test Case 3.2: Create Course - Missing Fields

Objective: Verify form validation
Test Steps:

1. Navigate to create course
2. Leave fields empty
3. Attempt submission

Expected Result:

- Form validation prevents submission
- Required fields highlighted

Status: ☐ Pass ☐ Fail

Test Case 3.3: Upload Lecture - Valid Video

Objective: Verify lecturer can upload video lectures
Prerequisites: At least one course exists

Test Steps:

1. Navigate to "Upload Lecture"
2. Enter lecture title: "Introduction"
3. Enter description: "Course introduction"
4. Select a course
5. Upload valid MP4 file (under 100MB)
6. Click "Upload Lecture"

Expected Result:

- File uploads successfully
- Video saved in static/uploads/videos/
- Database entry created with file path
- Success message displayed
- Lecture appears in course lecture list

Status: ☐ Pass ☐ Fail

Test Case 3.4: Upload Lecture - Invalid File Type

Objective: Verify system rejects invalid file types
Test Steps:

1. Navigate to upload lecture
2. Attempt to upload .txt or .exe file

Expected Result:

- Upload rejected
- Error message: "Invalid file type"

Status: ☐ Pass ☐ Fail

Test Case 3.5: Upload Lecture - File Too Large

Objective: Verify file size limit enforcement
Test Steps:

1. Attempt to upload file over 100MB

Expected Result:

- Upload rejected
- Appropriate error message

Status: ☐ Pass ☐ Fail

Test Case 3.6: Upload Lecture - No Course Created

Objective: Verify handling when no courses exist
Test Steps:

1. New lecturer with no courses
2. Navigate to upload lecture

Expected Result:

- Warning message displayed
- Link to create course first

Status: ☐ Pass ☐ Fail

Test Case 3.7: View Course Lectures

Objective: Verify lecturer can view their course lectures
Test Steps:

1. Navigate to "My Courses"
2. Click "View Lectures" on a course

Expected Result:

- All lectures for that course displayed
- Shows lecture details and upload dates
- Option to upload more lectures

Status: ☐ Pass ☐ Fail

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4. Student Module Tests

Test Case 4.1: View Available Courses

Objective: Verify student can browse courses
Test Steps:

1. Log in as student
2. Navigate to "Enroll in Course"

Expected Result:

- Available courses displayed
- Shows course descriptions and instructors
- "Enroll Now" button visible

Status: ☐ Pass ☐ Fail

Test Case 4.2: Enroll in Course

Objective: Verify student can enroll in courses
Test Steps:

1. Navigate to "Enroll in Course"
2. Click "Enroll Now" on a course
3. Confirm enrollment

Expected Result:

- Enrollment successful
- Success message displayed
- Course appears in "My Courses"
- Course removed from available courses list

Status: ☐ Pass ☐ Fail

Test Case 4.3: Duplicate Enrollment Prevention

Objective: Verify students cannot enroll twice
Test Steps:

1. Enroll in a course

2. Attempt to enroll again

Expected Result:

- Second enrollment rejected
- Appropriate error message

Status: ☐ Pass ☐ Fail

Test Case 4.4: View Enrolled Courses

Objective: Verify student can view enrolled courses
Test Steps:

1. Navigate to "My Courses"

Expected Result:

- All enrolled courses displayed
- Shows course details and lecture count
- "View Lectures" button available

Status: ☐ Pass ☐ Fail

Test Case 4.5: View Course Lectures

Objective: Verify student can view lectures for enrolled courses
Test Steps:

1. Navigate to "My Courses"
2. Click "View Lectures" on enrolled course

Expected Result:

- All lectures for course displayed
- Shows lecture titles and descriptions
- "Watch Lecture" button available

Status: ☐ Pass ☐ Fail

Test Case 4.6: Access Control - Non-enrolled Course

Objective: Verify students cannot access lectures from non-enrolled courses
Test Steps:

1. Find lecture ID for non-enrolled course
2. Try to access URL directly: /student/lecture/{id}

Expected Result:

- Access denied
- Error message: "Access denied. You are not enrolled in this course"
- Redirect to courses page

Status: ☐ Pass ☐ Fail

Test Case 4.7: Watch Video Lecture

Objective: Verify video streaming functionality
Test Steps:

1. Navigate to enrolled course lectures
2. Click "Watch Lecture"
3. Verify video player loads
4. Click play button

Expected Result:

- Video page loads
- Video player embedded
- Video plays successfully
- Controls available (play, pause, volume)
- Download option disabled

Status: ☐ Pass ☐ Fail

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5. Database Integration Tests

Test Case 5.1: Password Hashing

Objective: Verify passwords are hashed in database
Test Steps:

1. Create a new user
2. Check database directly

Expected Result:

- Password stored as hash, not plain text
- Hash starts with "scrypt:" or "pbkdf2:"

Status: ☐ Pass ☐ Fail

Test Case 5.2: Foreign Key Constraints

Objective: Verify referential integrity
Test Steps:

1. Attempt to delete a lecturer with courses
2. Check if courses are handled appropriately

Expected Result:

- Cascade delete or appropriate error
- Database maintains integrity

Status: ☐ Pass ☐ Fail

Test Case 5.3: Enrollment Unique Constraint

Objective: Verify unique enrollment constraint
Test Steps:

1. Enroll student in course
2. Attempt duplicate enrollment via direct database insert

Expected Result:

- Duplicate rejected by database
- Unique constraint error

Status: ☐ Pass ☐ Fail

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6. Security Tests

Test Case 6.1: SQL Injection Prevention

Objective: Verify system is protected against SQL injection
Test Steps:

1. In login form, enter: admin@dlms.com' OR '1'='1
2. Attempt various SQL injection patterns

Expected Result:

- Login fails
- No SQL errors exposed
- Parameterized queries prevent injection

Status: ☐ Pass ☐ Fail

Test Case 6.2: Session Security

Objective: Verify session cannot be hijacked
Test Steps:

1. Log in and note session cookie
2. Log out
3. Try to reuse old session cookie

Expected Result:

- Session invalidated after logout
- Old session cookie rejected

Status: ☐ Pass ☐ Fail

Test Case 6.3: File Upload Security

Objective: Verify malicious file uploads are prevented
Test Steps:

1. Attempt to upload .php or .exe file renamed as .mp4

2. Attempt to upload file with path traversal in name

Expected Result:

- Malicious files rejected
- Filename sanitized
- No code execution possible

Status: ☐ Pass ☐ Fail

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7. UI/UX Tests

Test Case 7.1: Responsive Design

Objective: Verify layout adapts to screen size
Test Steps:

1. View application on desktop (1920x1080)
2. View on tablet (768x1024)
3. View on mobile (375x667)

Expected Result:

- Layout adjusts appropriately
- All content accessible
- No horizontal scrolling

Status: ☐ Pass ☐ Fail

Test Case 7.2: Flash Messages

Objective: Verify user feedback messages
Test Steps:

1. Perform various actions (login, create user, etc.)
2. Verify appropriate messages appear

Expected Result:

- Success messages in green
- Error messages in red

- Warning messages in yellow
- Messages dismissible

Status: ☐ Pass ☐ Fail

Test Case 7.3: Navigation Flow

Objective: Verify intuitive navigation
Test Steps:

1. Navigate through all pages as each role
2. Verify back buttons work correctly

Expected Result:

- Clear navigation paths
- Back buttons return to appropriate pages
- No broken links

Status: ☐ Pass ☐ Fail

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8. Performance Tests

Test Case 8.1: Page Load Time

Objective: Verify acceptable page load times
Test Steps:

1. Clear browser cache
2. Load various pages
3. Measure load time

Expected Result:

- Pages load within 2 seconds
- Database queries optimized

Status: ☐ Pass ☐ Fail

Test Case 8.2: Video Streaming Performance

Objective: Verify smooth video playback
Test Steps:

1. Stream various video files
2. Check for buffering issues

Expected Result:

- Videos stream without excessive buffering
- Playback smooth for files under 100MB

Status: ☐ Pass ☐ Fail

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9. Integration Tests

Test Case 9.1: End-to-End Workflow - Course Creation to Viewing

Objective: Verify complete workflow
Test Steps:

1. Admin creates lecturer account
2. Lecturer logs in and creates course
3. Lecturer uploads video lecture
4. Admin creates student account
5. Student logs in and enrolls in course
6. Student views and watches lecture

Expected Result:

- All steps complete successfully
- Data flows correctly between components
- Video accessible to enrolled student

Status: ☐ Pass ☐ Fail

Test Summary Report

Test Execution Summary

- **Total Test Cases:** 45

- **Passed: ___**
- **Failed: ___**
- **Blocked: ___**
- **Not Executed: ___**

Critical Issues Found

1. _____
2. _____
3. _____

Recommendations

1. _____
2. _____
3. _____

Sign-off

- **Tested By: _____**
- **Date: _____**
- **Status:** ☐ Approved ☐ Rejected ☐ Conditional Approval

Notes for Testers

- Always test with fresh database for consistent results
- Document any unexpected behavior
- Take screenshots of errors
- Note browser and OS version for any issues
- Verify all test prerequisites before starting
- Report security issues immediately